

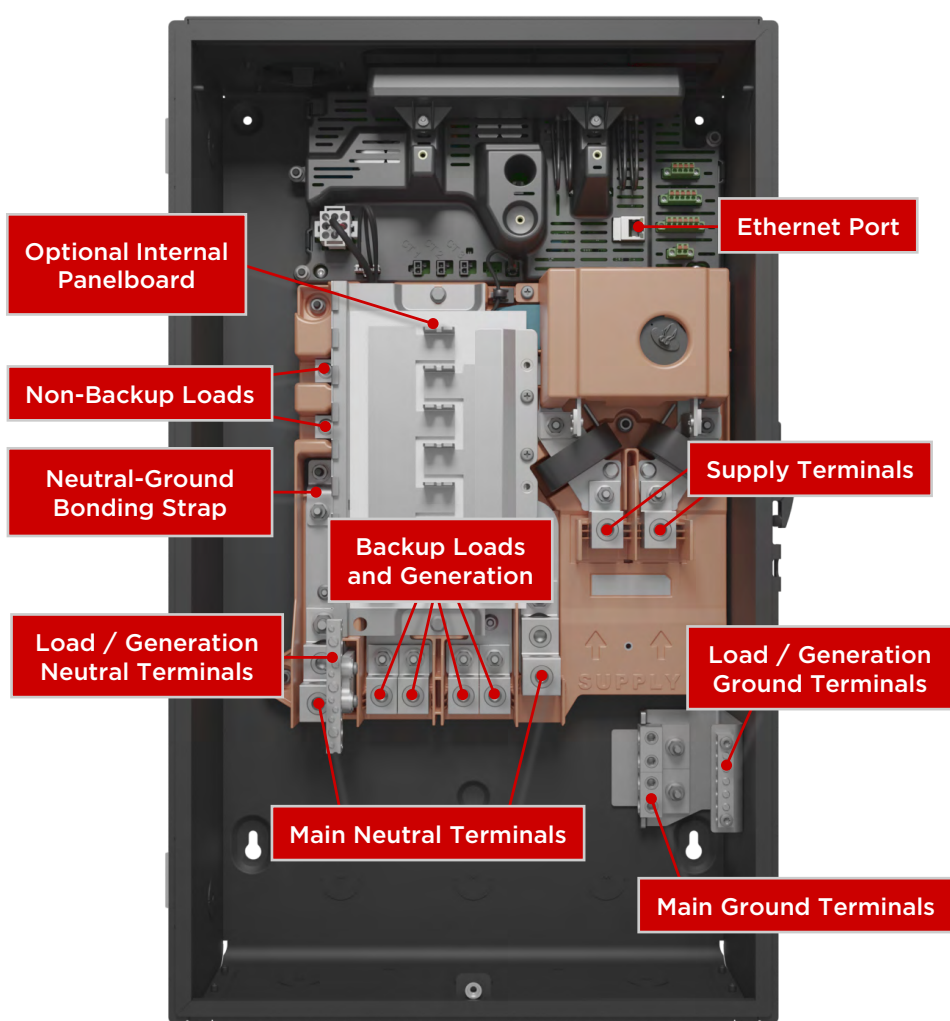


POWERWALL WITH BACKUP GATEWAY 2

[View Partner Portal](#)

The *Powerwall 2 AC (Backup Gateway 2) Installation Manual (North America)* can be found in the *Installation* section in the Powerwall documents

BACKUP GATEWAY 2 OVERVIEW



SYSTEM WIRING CONFIGURATIONS

Backup Gateway
Without Internal Panelboard



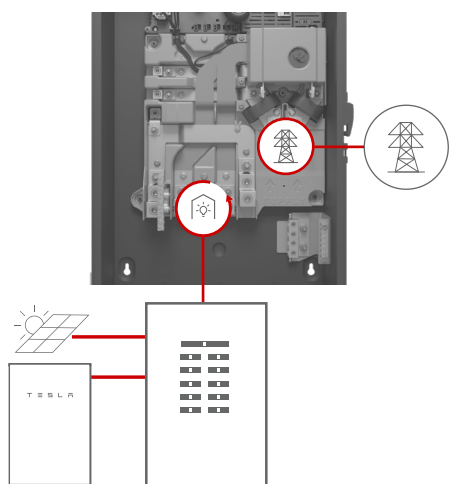
Backup Gateway
With Internal Panelboard



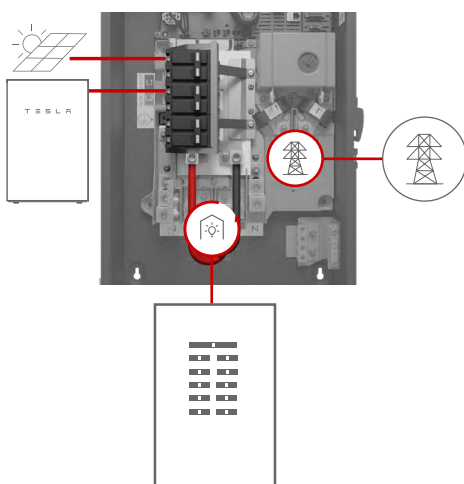
WHOLE HOME BACKUP

Where all loads can be supported by Powerwall

Powerwall and Solar
in existing panel



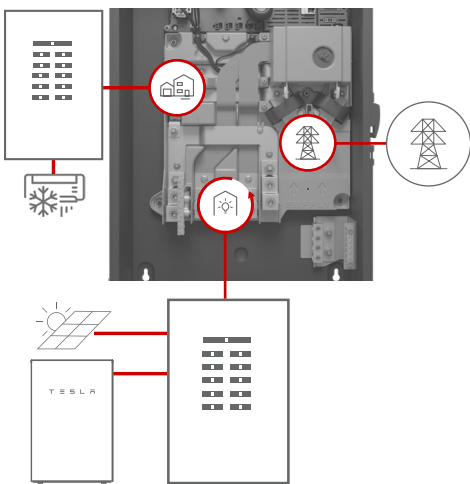
Powerwall and Solar
on backup internal panelboard



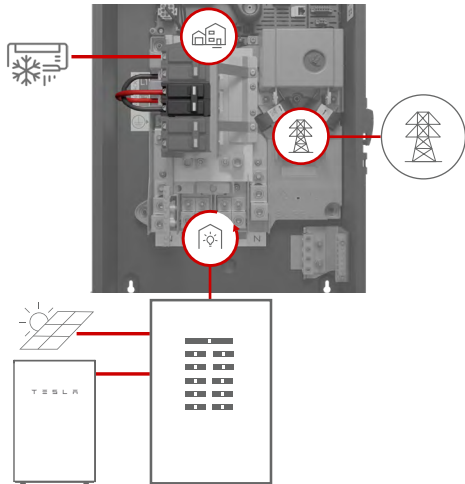
PARTIAL HOME BACKUP

Heavy loads cannot be supported by Powerwall

Heavy loads in existing non-backup subpanel



Heavy loads on non-backup internal panelboard



See **System Wiring Diagrams** in the *Installation Manual* for detailed diagrams of common configurations

LEGEND



Backup Loads



Non-Backup Loads



High Demand Loads

REQUIRED TOOLS

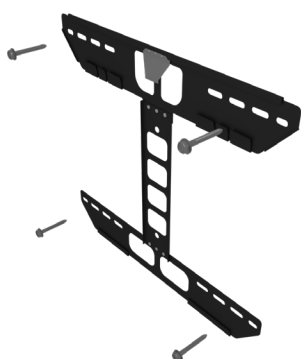


SUGGESTED TOOLS

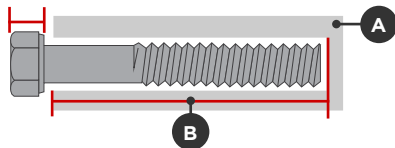


MOUNT POWERWALL

1



Max 9 mm

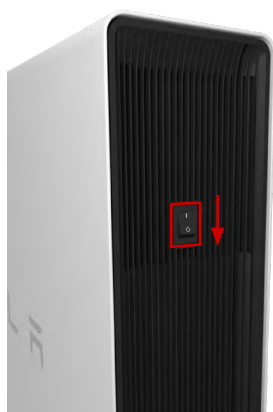


A	Material	B	Minimum Embedment
	Wood Studs		2 1/2"
	Plywood		1/4" beyond back of plywood (min 1/2" plywood)
	Metal Studs		3 threads beyond stud / backing
	Concrete / Masonry		1 1/2"
	Channel Strut		2 1/2" into wood studs 3 threads beyond metal studs



See **Powerwall Mounting Details** in the *Installation Manual* for additional details

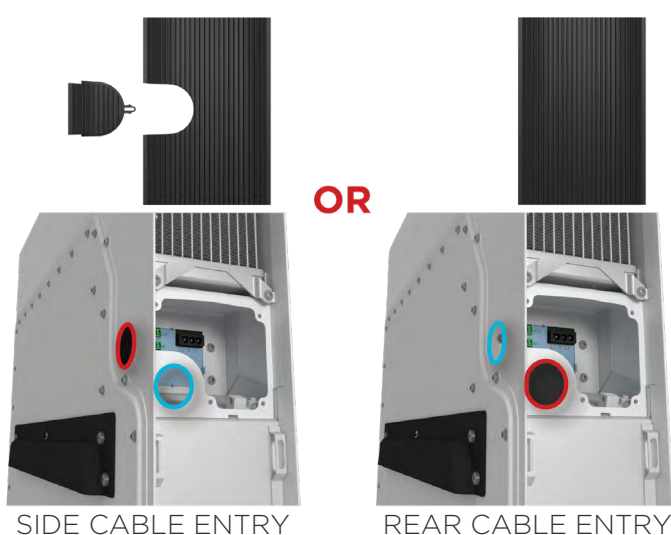
2



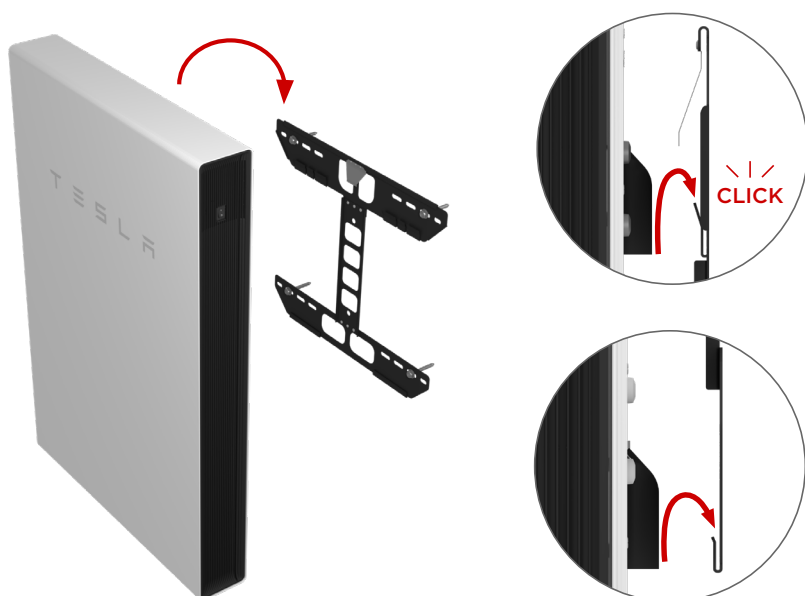
3



4



5



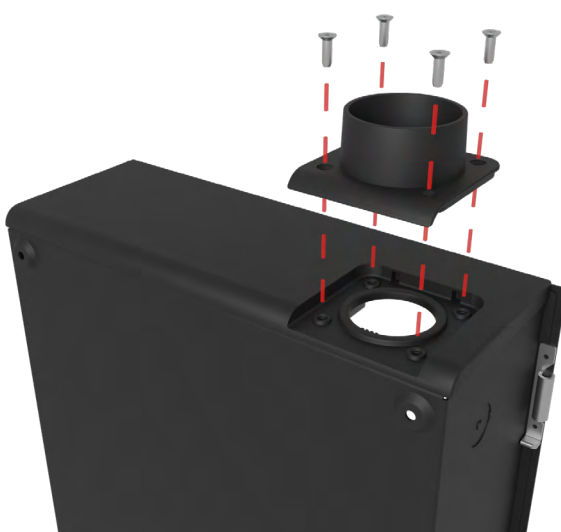
MOUNT THE BACKUP GATEWAY

6

BOTTOM, REAR, OR SIDE CABLE ENTRY



OPTIONAL: TOP CABLE ENTRY VIA CONDUIT HUB KIT



7

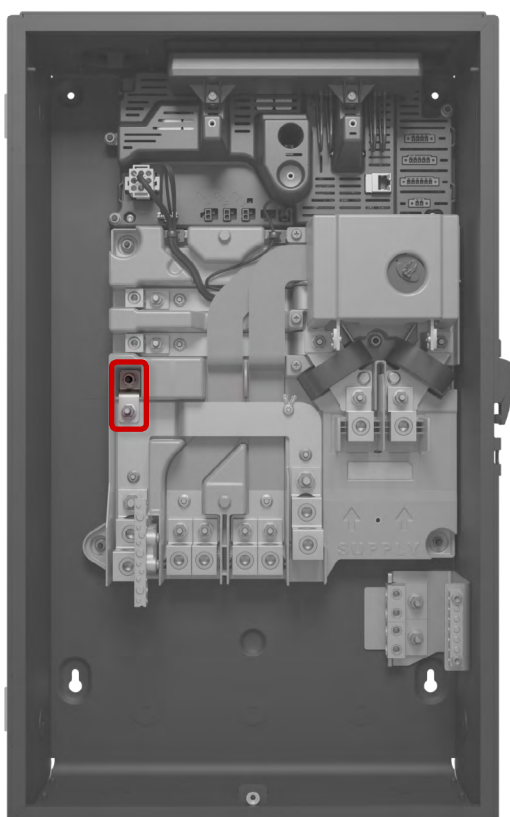


8

IS GATEWAY INSTALLED AS SERVICE EQUIPMENT?

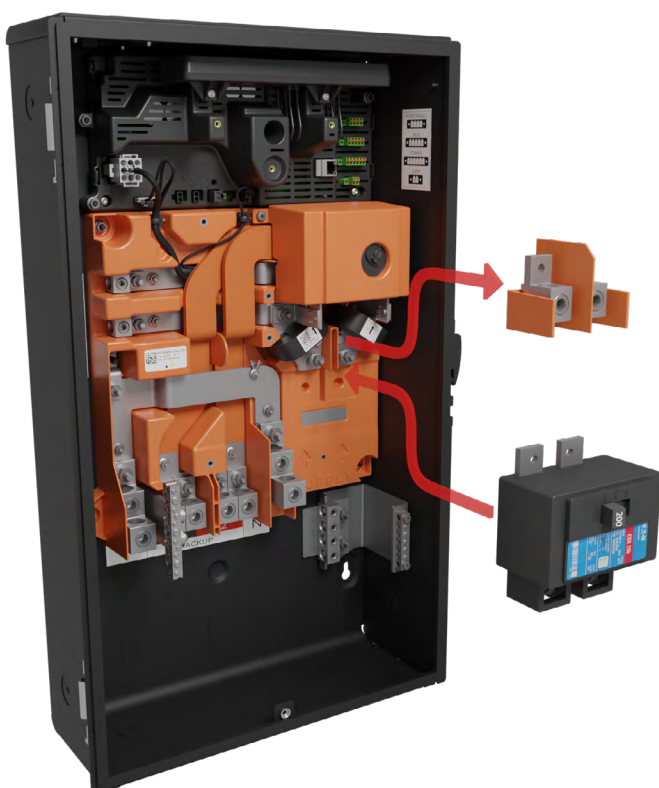
NO

GATEWAY **IS NOT** INSTALLED AS SERVICE EQUIPMENT
REMOVE NEUTRAL-GROUND BONDING STRAP



YES

GATEWAY **IS** INSTALLED AS SERVICE EQUIPMENT
INSTALL MAIN BREAKER

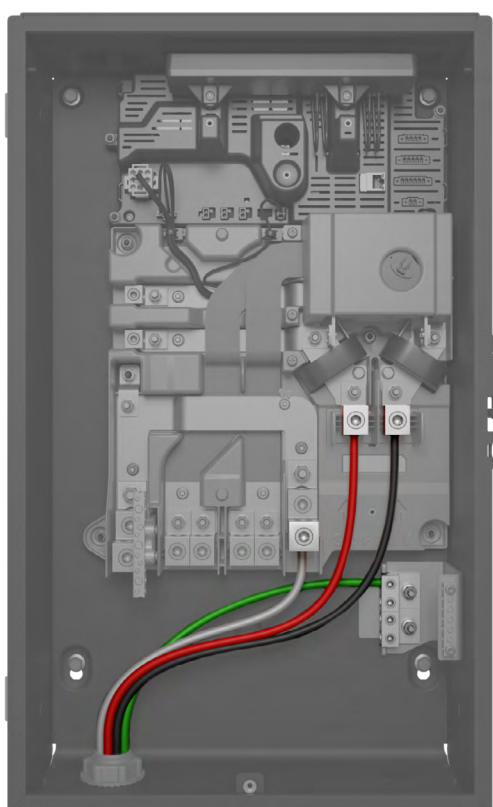


See **Wiring Reference** in the *Installation Reference Manual* for breaker types

MAKE BACKUP GATEWAY AC POWER CONNECTIONS

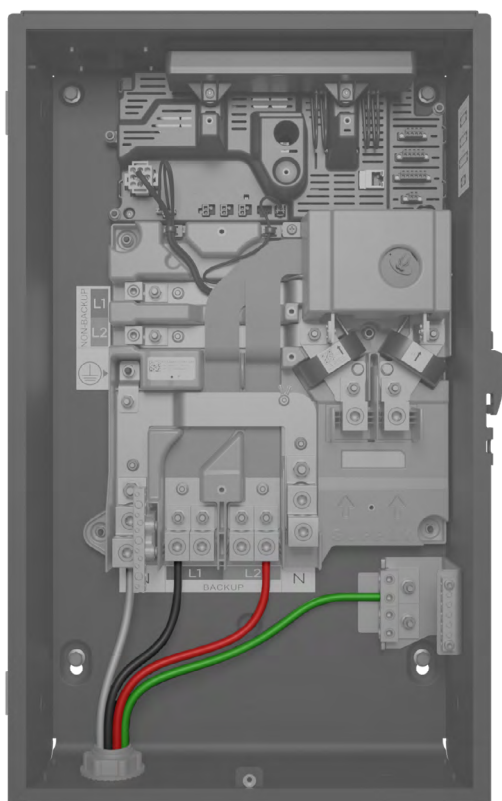
9

SUPPLY SIDE CONDUCTORS



- 1 1/4"
- 6 AWG - 250 kcmil
- 6 AWG - 250 kcmil
- 6 AWG - 250 kcmil
- 275 in-lbs
- 5/16"
- 3/4"
- 14 AWG - 2/0
- 120 in-lbs
- 3/16"

BACKUP LOAD PANEL CONDUCTORS



6 AWG - 250 kcmil   

275 in-lbs



5/16"

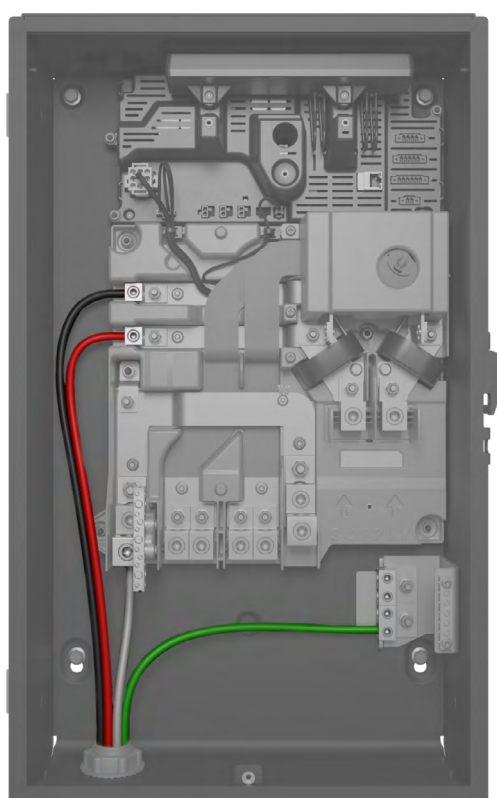
14 AWG - 2/0 

120 in-lbs



3/16"

NON-BACKUP PANEL CONDUCTORS



6 AWG - 250 kcmil   

275 in-lbs



5/16"

14 AWG - 2 AWG  

14 AWG - 2 AWG 

14 AWG - 2/0 

120 in-lbs



3/16"

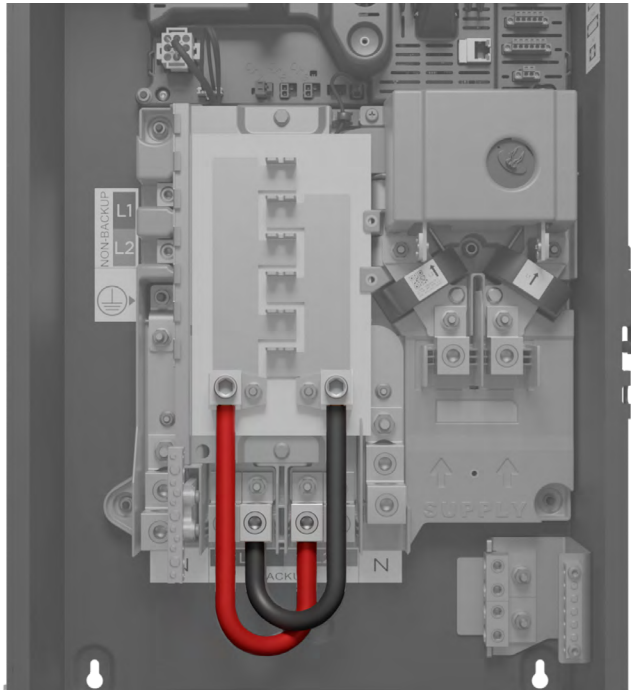
OPTIONAL: INSTALL INTERNAL PANELBOARD

10



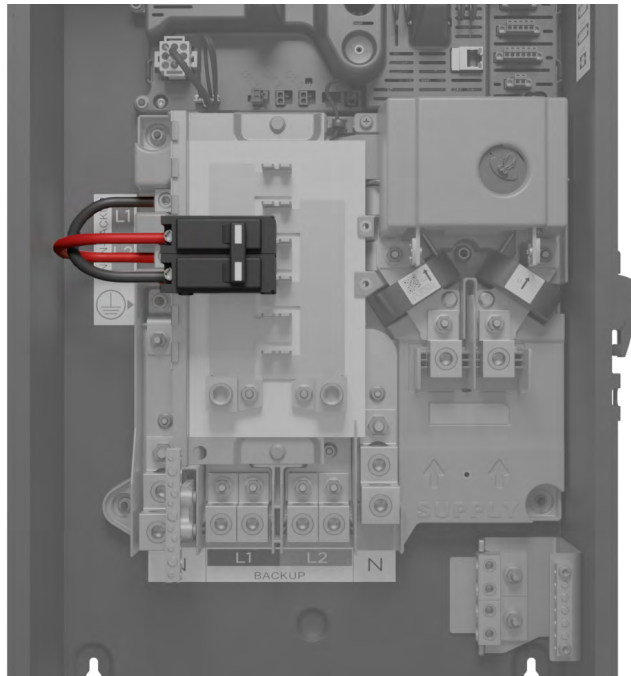
11

BACKUP INTERNAL PANELBOARD



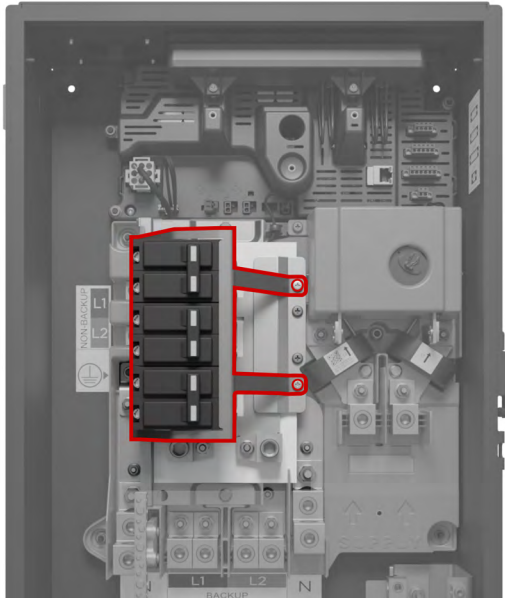
OR

NON-BACKUP INTERNAL PANELBOARD



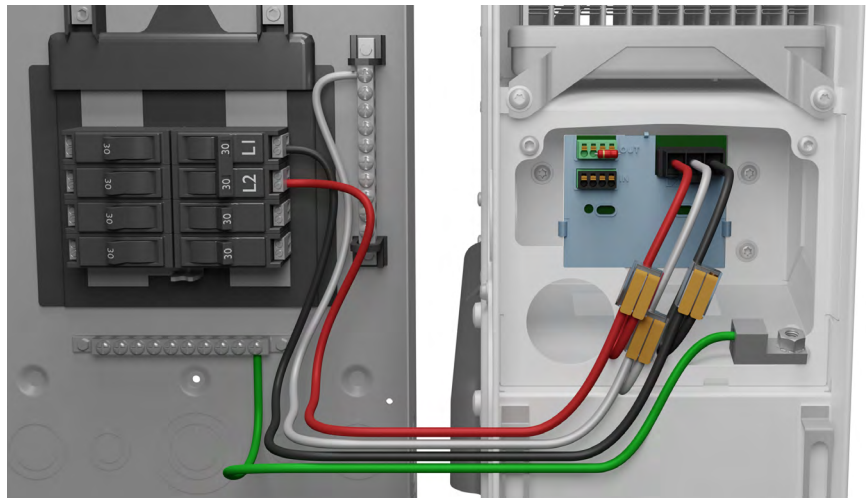
12

INSTALL BREAKER HOLD-DOWN



MAKE POWERWALL AC POWER CONNECTIONS

13

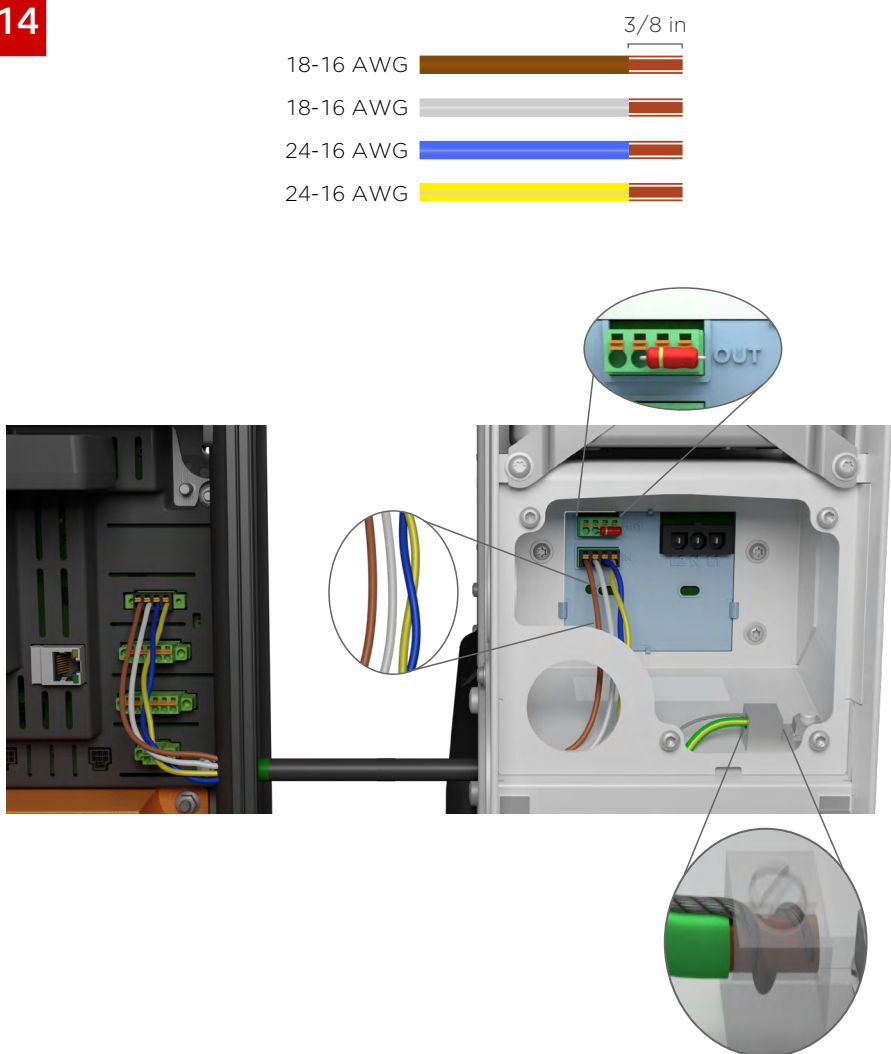


7/16"

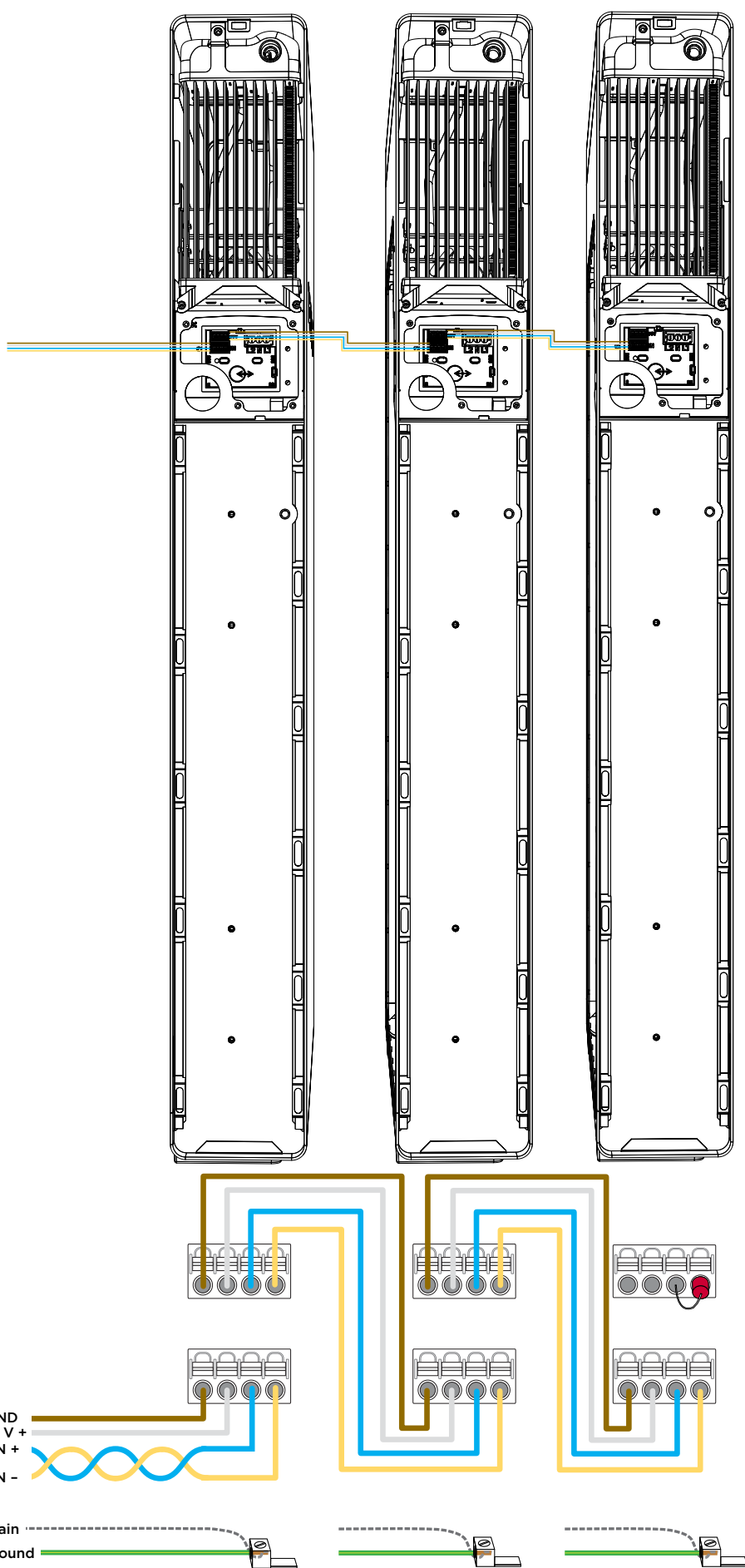
10 - 8 AWG  

10 - 8 AWG  

10 - 8 AWG  



MULTI-POWERWALL COMMUNICATION WIRING



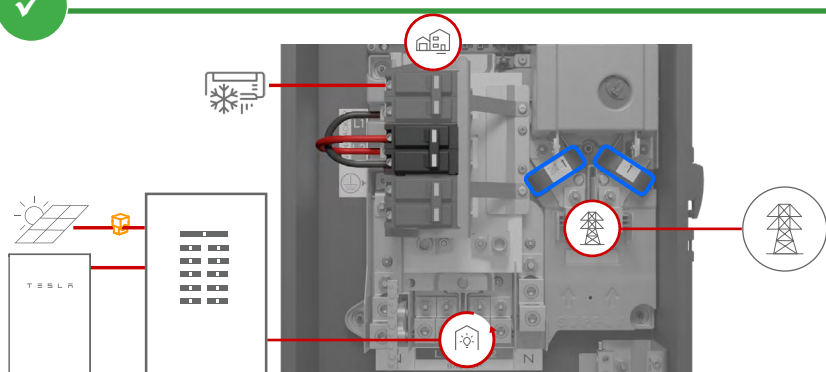


See **Install Optional Remote Disable Switch** in the *Installation Manual* for information on switch operation and design considerations.

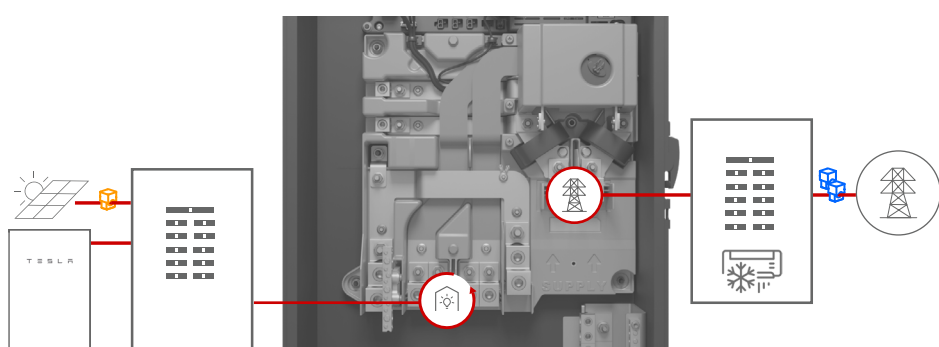
INSTALL METERING

METERING OVERVIEW

Correct Metering Installation



All **Site** loads monitored with two integrated CTs 
All **Solar** monitored with one auxiliary CT 

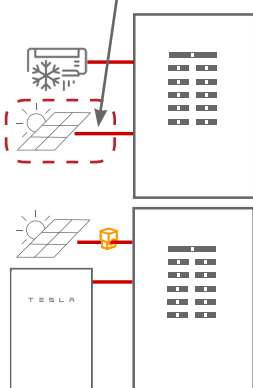


All **Site** loads monitored with two auxiliary CTs 
All **Solar** monitored with one auxiliary CT 

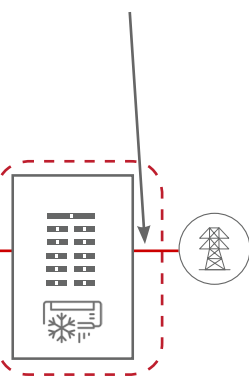


Incorrect Metering Installation

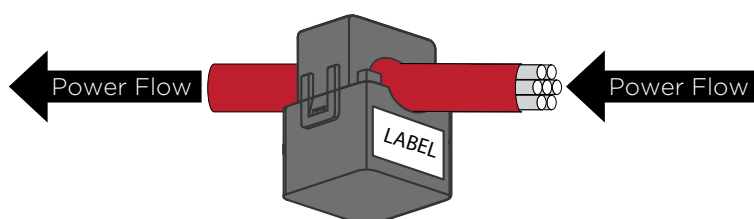
Secondary **Solar** is **not** monitored



Non-Backup **Site** loads are **not** monitored by integrated CTs



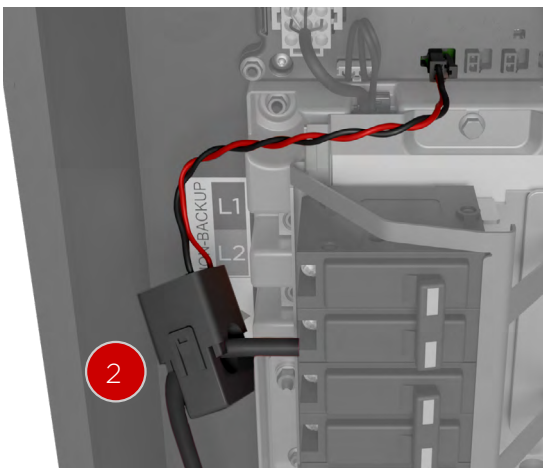
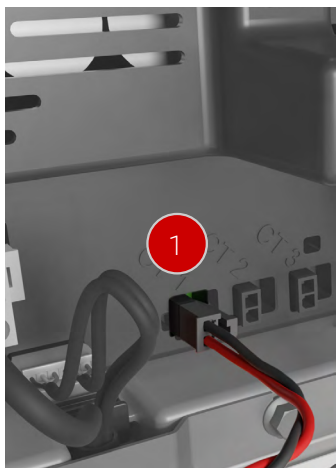
CT ORIENTATION



CT Label → Source of Power

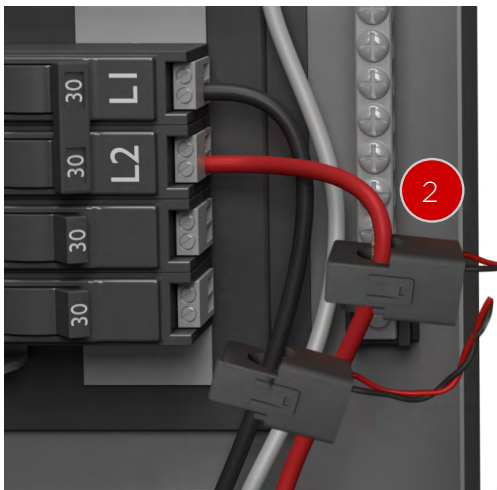
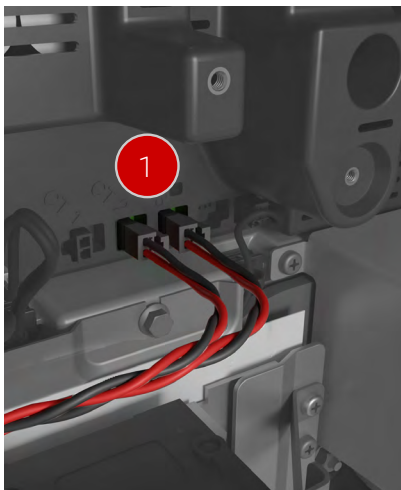
16

INSTALL SOLAR METERING



INSTALL REMOTE SITE METERING

(Required for Home Loads upstream of Gateway)



Note: If installing a Neuroio meter and additional CTs, connect the Neuroio over Wi-Fi to update

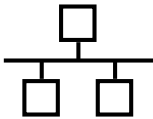


See the *Installation Manual* for a detailed overview of metering, system wiring diagrams with meter placement, and instructions for installing a remote Neuroio meter.

CONNECT BACKUP GATEWAY TO THE INTERNET

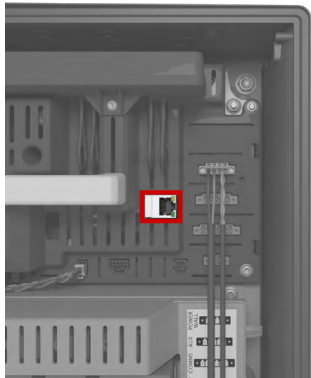
17

Ethernet



AND / OR

2.4 GHz / 5GHz Wi-Fi



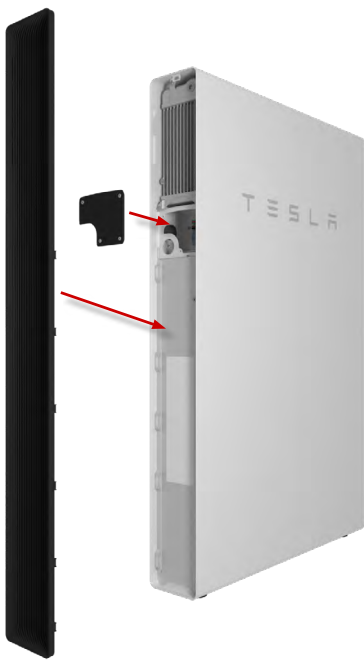
HOMEOWNER NETWORK
HOMEOWNER PASSWORD

CLOSE WIRING COMPARTMENTS AND POWER ON

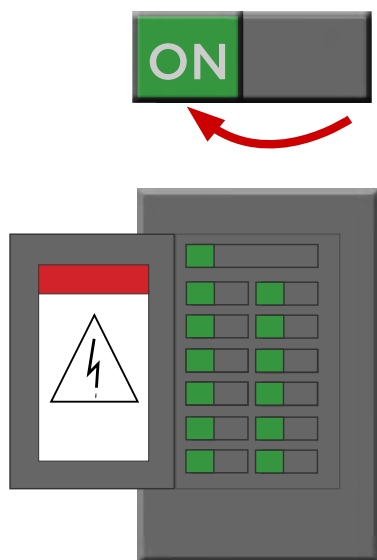
18



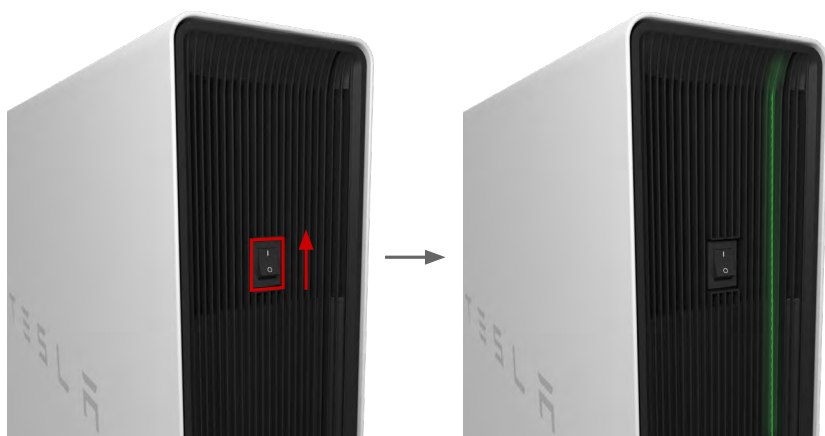
19



20



21



COMMISSION THE SYSTEM

22



23



TIPS

- Use **Chrome: Incognito** mode or **Safari: Private Browsing** mode.
- Work in one browser tab only.
- For security message **Your connection is not private**, select **Advanced** and **Proceed**.
- For **Error: Bad Credentials**, select **FORGOT PASSWORD** to reset password.
- For **Error: Login Error (Cannot Start Wizard)**, select **Click for More** and tick the **Force Launch Wizard** option. Try again.
- If experiencing other issues, refresh the web page.
- If issues persist, reboot Gateway using **RESET** button.